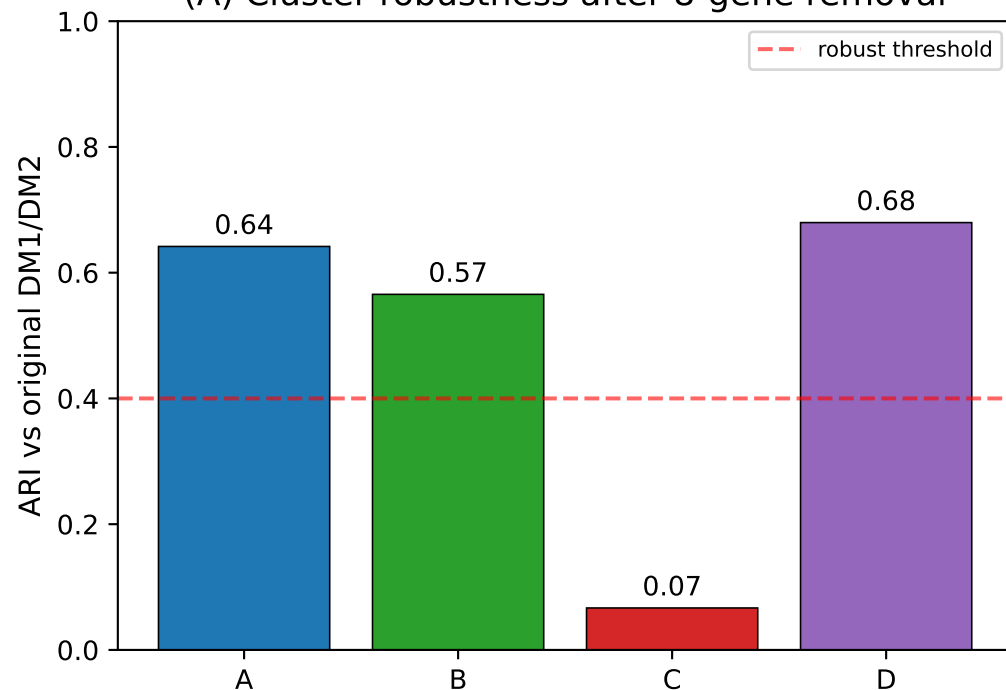
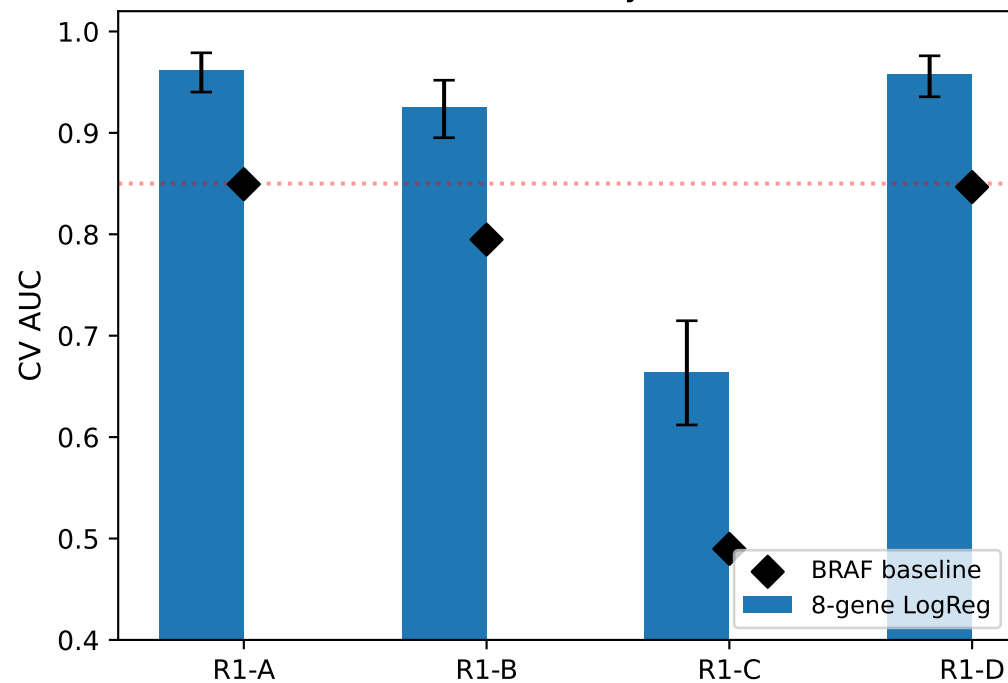


v17 REAL FIX summary — circular validation flaw mitigated; 8-gene panel is genuine biomarker

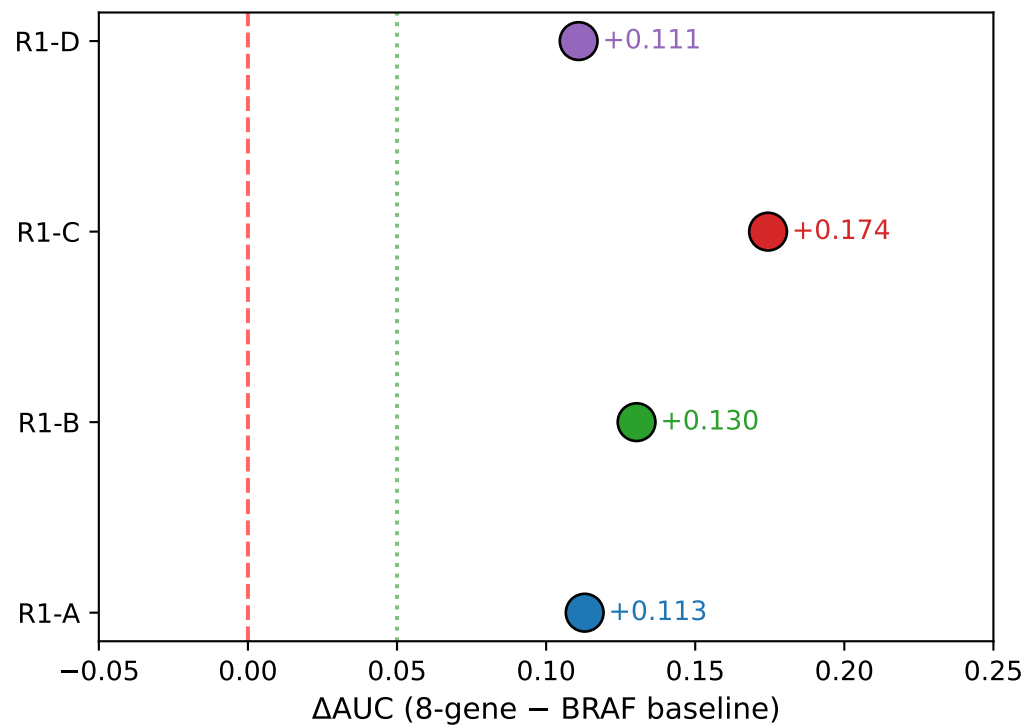
(A) Cluster robustness after 8-gene removal



(B) Leak-free CV AUC by cluster variant



(C) Δ AUC over BRAF — sustained at +0.11~+0.17 across all variants



(D) Pre-fix vs Post-fix — performance preserved

